NUR E JANNAT ASHA

AI & Machine Learning Engineer | GenAI, LLMs & High-Performance Systems

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github.com/Jannat332 jannat332.github.io/portfolio linkedin.com/in/nur-e-jannat-asha

✓ Full Finnish Work Rights (LUT M.Sc. Graduate) ✓ Available Immediately • Espoo / Helsinki / Remote

★ Eligible for Aalto University F.E.C. Subsidized Recruitment Traineeship English (Fluent) • Finnish (A1 – Active)

42% Hallucination ↓	60% GPU Speedup	4.24 / 5.00	5 Publications
RAG + KG CONFIDENCE FUSION	CUDA / C++ ACCELERATION	LUT M.SC. DISTINCTION	IEEE & ELSEVIER JOURNALS

PROFESSIONAL SUMMARY

AI & Machine Learning Engineer with an M.Sc. in Software Engineering (**4.24/5.00 CGPA, Distinction**) from LUT University, Finland. Specialized in building reliable Generative AI systems, **Retrieval-Augmented Generation (RAG)**, knowledge graph validation, and GPU-accelerated computing (CUDA). Authored 5 peer-reviewed scientific publications (IEEE and Elsevier) in numerical simulation, signal processing, and computational modeling. Experienced in bridging research prototypes to production microservices using Python, FastAPI, Docker, and automated CI/CD. Available immediately for full-time AI engineering positions or placement through the **Aalto University F.E.C.** recruitment program.

TECHNICAL EXPERTISE

AI, ML & GenAI:	**Retrieval-Augmented Generation (RAG)**, **Knowledge Graph Validation**, Prompt Engineering, Hallucination Mitigation, Embeddings & Vector Search, Confidence Fusion Scoring
Languages & Frameworks:	**Python (3.10+, FastAPI, PyTorch, NumPy, Pandas, Scikit-Learn)**, **C++17**, **CUDA**, SQL (PostgreSQL, SQLite), Bash, React 18
High-Performance Computing:	GPU Kernel Parallelization, Numerical Simulation, Matrix Computation, Cache Optimization, Asynchronous Processing
DevOps & MLOps:	**Docker** (reproducible environments), **GitHub Actions CI/CD**, Linux/Unix, Git, RESTful API Design, Postman
Scientific & Analytical:	Mathematical Modeling, Statistical Error Analysis, Signal Acquisition & Noise Filtering (PPG / IoT Sensors), TDD (PyTest, GoogleTest)
Languages:	**English** (Fluent / Full Professional), **Finnish** (A1 – actively studying weekly), Bengali (Native)

FEATURED AI & PRODUCTION REPOSITORIES

Verify code directly at github.com/Jannat332

LLM Parallel Reasoning & Verification Engine `Python 3.11` `RAG` `Knowledge Graph` github.com/Jannat332/llm-parallel-reasoning ↗

Dual-Stream Semantic Retrieval & Dynamic Contradiction Scoring

- Designed a dual-stream reasoning architecture fusing dense vector retrieval with explicit knowledge-graph relational verification, **reducing factual hallucinations by 42%** on complex multi-hop queries.
- Implemented a confidence-scoring arbitration engine that quantifies semantic entropy and validates assertions before presenting generated output to users.
- Packaged benchmark evaluation suites and CLI runners inside containerized Docker images for deterministic, reproducible execution.

Quantitative Stock Signal & Technical Alert Engine `Python` `Algorithms` `Financial ML` github.com/Jannat332/stock-signal-alerts ↗

Automated Market Momentum Indicators & Signal Trigger Pipeline

- Implemented mathematical technical indicator algorithms (SMA 20/50 Crossover, 14-period RSI, Momentum Divergence) with zero third-party financial calculation library dependency.
- Built an event-driven alert dispatcher generating machine-readable buy/sell signals with automated trend validation and threshold checking.
- Integrated comprehensive PyTest test suites verifying boundary conditions, zero-division safeguards, and extreme market volatility regimes.

Deterministic Embedded Traffic Controller `C++17` `FSM` `Safety Invariants` github.com/Jannat332/traffic-light-controller ↗

Edge AI & Safety-Critical Embedded State Machine

- Engineered deterministic state machine with thread-safe preemption and zero race conditions in modern C++17.
- Enforced 100% safety invariants with GoogleTest suite verifying hazard transitions and pedestrian walk timings.

Automated CI/CD Pipeline & Docker Security Template `GitHub Actions` `MLOps` github.com/Jannat332/github-actions-ci-template ↗

Automated Testing, Multi-Stage Containerization & Quality Gatekeeping

- Constructed a GitHub Actions workflow automating linting, security vulnerability scanning, and multi-stage Docker builds with a **65% image footprint reduction**.

PROFESSIONAL EXPERIENCE

Research Assistant (Computational Systems) — North South University 2023 – 2024
GPU Acceleration, Numerical Modeling & Simulation Workflows

- Accelerated scientific simulation algorithms by **60% using CUDA GPU kernel parallelization** and vectorization in C++ and Python.
- Engineered automated data preprocessing, numerical convergence checking, and statistical validation workflows for complex physical dynamics.
- Co-authored 3 computational research publications in prestigious international Elsevier scientific journals.

Junior Software Engineer — Spectrum IT Solutions 2021 – 2022
Backend Engineering, Service Optimization & Linux Systems

- Developed and maintained production backend APIs and data endpoints in Python and C++ for enterprise clients.
- Conducted performance profiling and debugging with GDB and Valgrind, improving data query response latency by 25%.
- Collaborated in Agile sprints using Git workflows, automated test pipelines, and peer code reviews.

EDUCATION & QUALIFICATIONS

Master of Science in Software Engineering and Digital Transformation — LUT University, Finland 2024 – 2025
CGPA: 4.24 / 5.00 (Distinction • Top 5% of Graduating Class)
Curriculum: Advanced Software Architecture, Distributed Data Systems, Cloud Computing, Modern Engineering Practices.

Bachelor of Science in Computer Science and Engineering — North South University 2016 – 2021
CGPA: 3.44 / 4.00 • Embedded sensor system and physiological signal classification published in **IEEE Xplore**.

Aalto University F.E.C. (Further Educated with Companies) Program — Aalto University, Espoo, Finland Intake 2025 – 2026
Accepted participant. Provides Finnish host companies with a low-risk, subsidized 6-month recruitment traineeship transitioning into a permanent software engineering contract.

SELECTED SCIENTIFIC PUBLICATIONS

- **IEEE ICOEI 2021:** *"Design of Low-Cost Sensor Embedded System for Real-Time Physiological Signal Processing."* DOI: 10.1109/ICOEI51242.2021.9452873
- **Elsevier International Journal of Thermofluids (2024):** Numerical modeling and computational analysis of thermal transport dynamics. DOI: 10.1016/j.ijft.2024.100650
- **Elsevier Heliyon (2024):** Computational modeling of fluid-structure thermal phenomena. DOI: 10.1016/j.heliyon.2024.e32306
- **Elsevier Case Studies in Thermal Engineering (2024):** Finite element simulation and parametric optimization. DOI: 10.1016/j.csite.2024.104273